PERFECT PAIRINGS FOR A PERFECT COMPLEX AGAINST THE DUALIZING COMPLEX OF A PROPER SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVY (sources/stacks-duality/pinned-original-source.tex line 7078).

Allowed standard prerequisites (editorial).

- Duality for a proper scheme over a field. Same chapter, Lemma labelled lemma-duality-proper-over-field, cited at proof lines 7097-7099. Fixes k, X and the dualizing complex.
- The trace pairing on a proper scheme over a field. Same chapter, Remark labelled remark-duality-proper-over-field, cited at proof lines 7097-7103. Contains the pairing whose perfectness is invoked.
- Composition of Ext groups is given by the cup product. Cohomology, Lemma labelled cohomology-lemma-ext-composition-is-cup, cited at proof lines 7099-7101. Identifies the displayed pairing with the one of the remark.

Source reference. Original lemma 0FVV: sources/stacks-duality/pinned-original-source.tex, source lines 6934–6934. Same-chapter pointer into the author environment lemma at source line 6934; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FVW: sources/stacks-duality/pinned-original-source.tex, source lines 7011–7011. Same-chapter pointer into the author environment remark at source line 7011; invoked at proof line 7101 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0FKU: sources/stacks-chow/tags.txt, source lines 19474–19474. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is

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Source reference. Original section 0FPC: sources/stacks-chow/tags.txt, source lines 19597–19597. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FVB: sources/stacks-chow/tags.txt, source lines 19806–19806. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7099 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k, X, and $\omega_X^\bullet$ be as in Lemma 0FVV. Let $t : H^0(X, \omega_X^\bullet) \to k$ be as in Remark 0FVW. Let $E \in D(\mathcal{O}_X)$ be perfect. Then the pairings*

$$H^i(X, \omega_X^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_X} E^\vee) \times H^{-i}(X, E) \longrightarrow k, \quad (\xi, \eta) \longmapsto t((1_{\omega_X^\bullet} \otimes \epsilon)(\xi \cup \eta))$$

are perfect for all i. Here $\cup$ denotes the cup product of Cohomology, Section 0FKU and $\epsilon : E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E \to \mathcal{O}_X$ is as in Cohomology, Example 0FPC.

Proof. By replacing E with $E[-i]$ this reduces to the case $i = 0$. By Cohomology, Lemma 0FVB we see that the pairing is the same as the one discussed in Remark 0FVW whence the result by the discussion in that remark. □